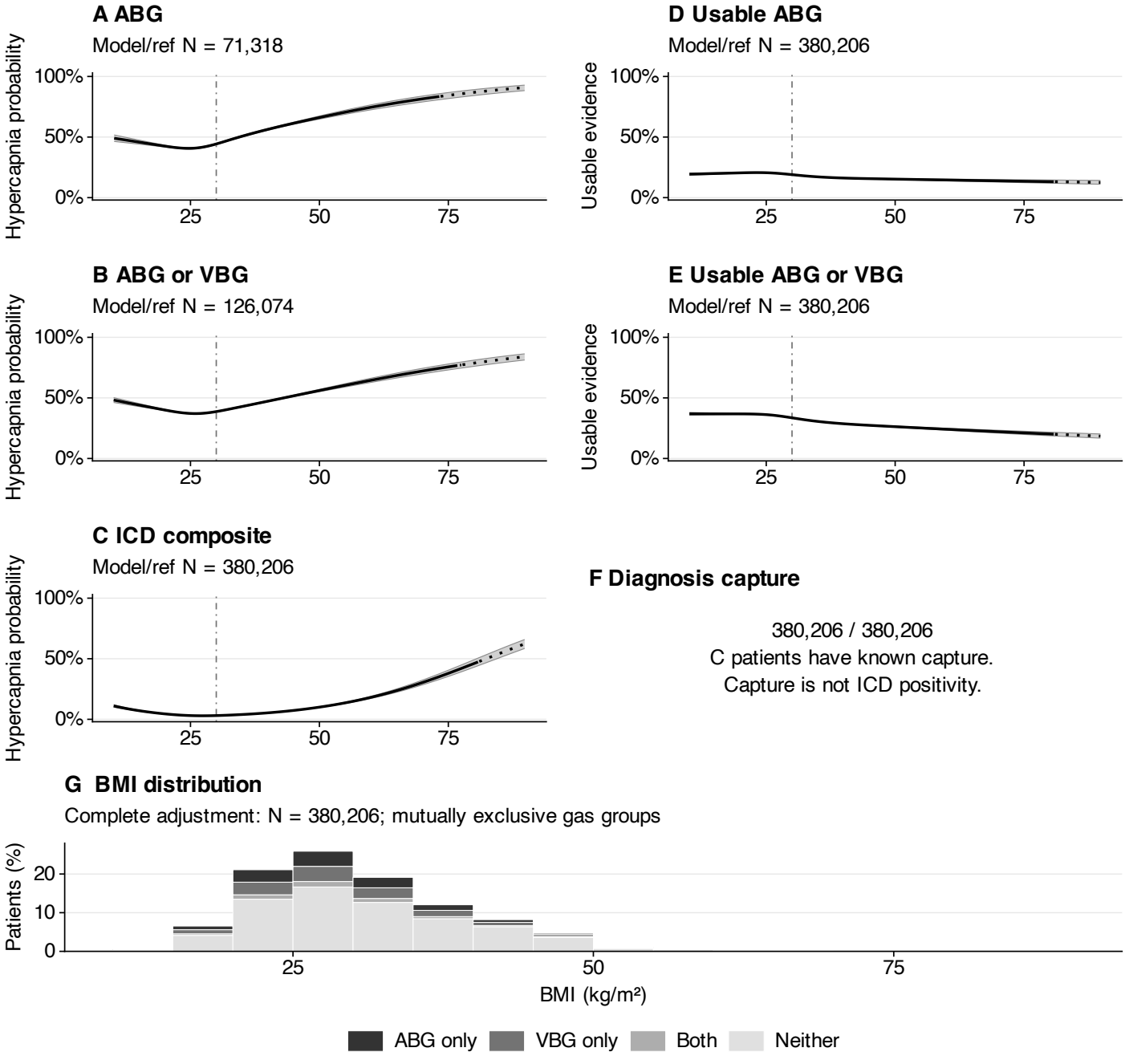
BMI and documented hypercapnia

All valid BMI: primary | encounter-wide



Shading: 95% CI | Dotted: locally sparse | Dashed: outside training BMI range

primary/F1-B

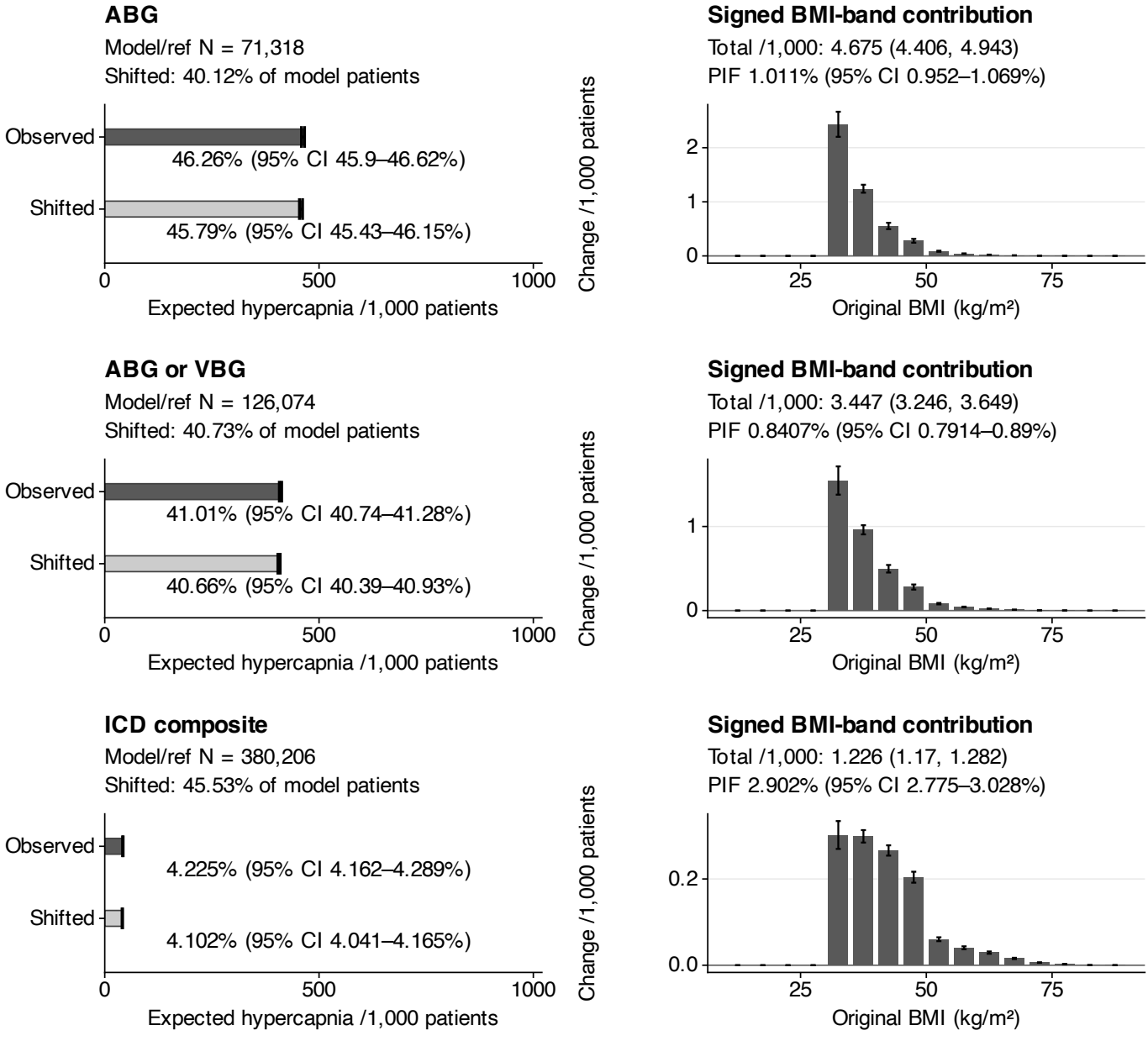
Analysis population: All valid BMI: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F1. BMI–hypercapnia relationships at the first eligible adult 2022 acute-care encounter. ABG denotes $PaCO_2$ ≥45 mm Hg; ABG or VBG denotes $PaCO_2$ ≥45 or $PvCO_2$ ≥50 mm Hg. ICD denotes the five-code composite, including E66.2. The vertical dot-dash line marks BMI 30, the scenario eligibility boundary, not the cohort boundary. Gas curves standardize over outcome-evaluable patients; ICD uses known diagnosis capture. Model/ref N is the common model and reference count. Availability and BMI frequency use all adjustment-complete patients (C); the four frequency groups are mutually exclusive. Adjustment: age, sex, race/ethnicity and region with a fixed natural BMI spline. Shading and fine boundaries show pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. Open availability points use Wilson intervals when no model is estimable. Known capture is distinct from ICD positivity. ABG/VBG: arterial/venous blood gas. Conditional model uncertainty excludes selection and measurement assumptions; no population transport or causal effect is inferred.

New results: investigator interpretation and figure approval pending.

One-unit BMI reduction for observed BMI >30; no floor

Paired predictions; original reference and covariates retained

All valid BMI: primary | encounter-wide



primary/F2

Analysis population: All valid BMI: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F2. Paired scenario: One-unit BMI reduction for observed BMI >30; no floor. ABG: arterial PCO_2 ≥45 mm Hg; ABG or VBG: arterial ≥45 or venous ≥50 mm Hg; ICD: five-code composite including E66.2. Left: observed and shifted standardized probabilities with model-based 95% CIs and printed limits. Left axes share the full 0–1,000 reference-patient scale; printed probabilities and limits retain small differences. N identifies the common model/reference population; shifted percentage is a descriptive cohort fraction. Right: signed original-BMI-band contributions per 1,000 reference patients, with 95% CIs. Patients at BMI ≤30 remain in baseline burden and contribute exactly zero scenario difference; the eligible share is printed. Total-difference/PIF limits are printed; contribution covariance reproduces total variance. Right axes vary by outcome and retain zero and all limits. Gas samples are outcome-evaluable; ICD uses known capture. Adjustment, source-entry membership and covariates are fixed. PIF is the model-standardized potential impact fraction. These conditional scenarios do not estimate a weight-loss treatment effect.

New results: investigator interpretation and figure approval pending.

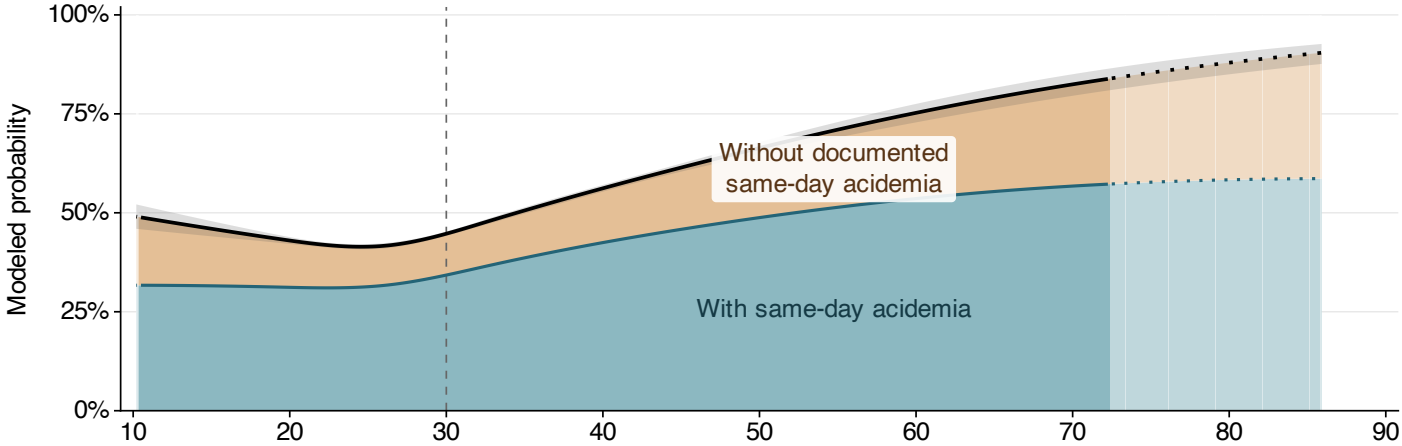
BMI and hypercapnia with and without same-day acidemia

Same recorded-day co-occurrence; not verified same-sample pairing

All valid BMI: primary; model/ref N = 53,607; with acidemia: 18,801; without: 6,200

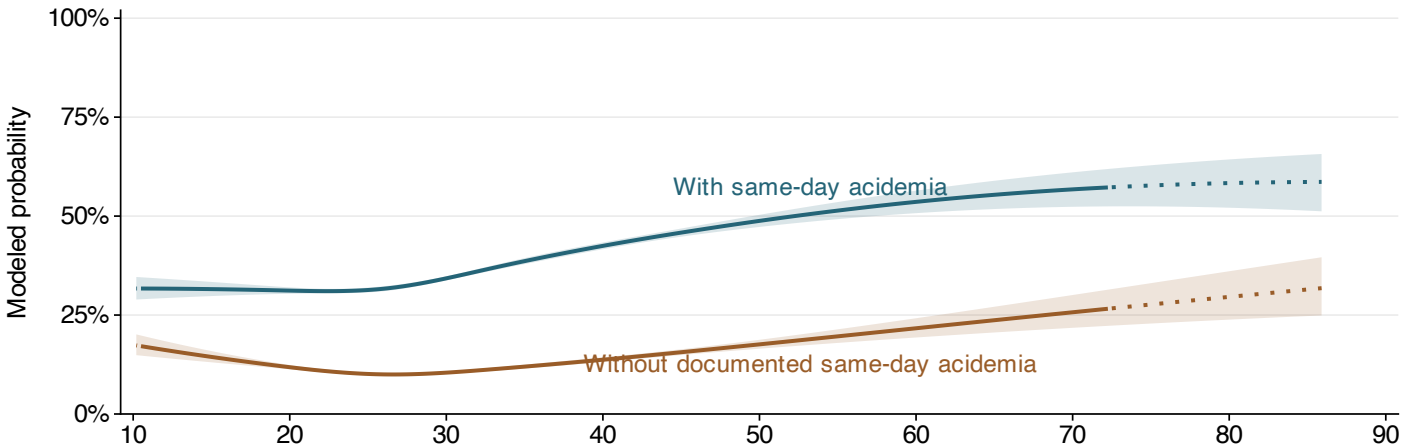
A Total hypercapnia, divided into its two components

Band thickness = component probability; top boundary = total (95% CI)

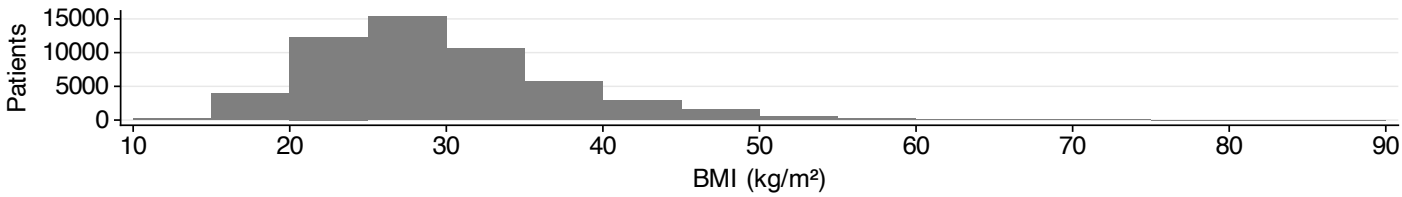


B Component curves on a common baseline

Pointwise 95% intervals; dotted segments mark sparse local evidence



C Patients in the shared co-observed population



primary/S1-same-day-ABG

All valid BMI: primary. Same recorded-day co-occurrence; not verified same-sample pairing. Fixed-index patients with both arterial analytes share inherited adjustment and their analysis population's fixed BMI spline. Any eligible day with $PaCO_2$ ≥45 mm Hg and pH <7.35 assigns the with-acidemia category; remaining hypercapnic patients have no documented same-day acidemia. Different samples may contribute. A: band thicknesses give component probabilities; the upper boundary is total hypercapnia. B: components share a baseline. Shading denotes pointwise 95% intervals using full joint covariance. C: shared-population BMI frequencies. Fading/dotted lines mark local windows below 30 patients or five events/non-events; marginal checks do not establish joint support. These recorded outcomes do not establish simultaneous physiology, acute respiratory failure or compensation. Selection, treatment timing and confounding remain limitations.

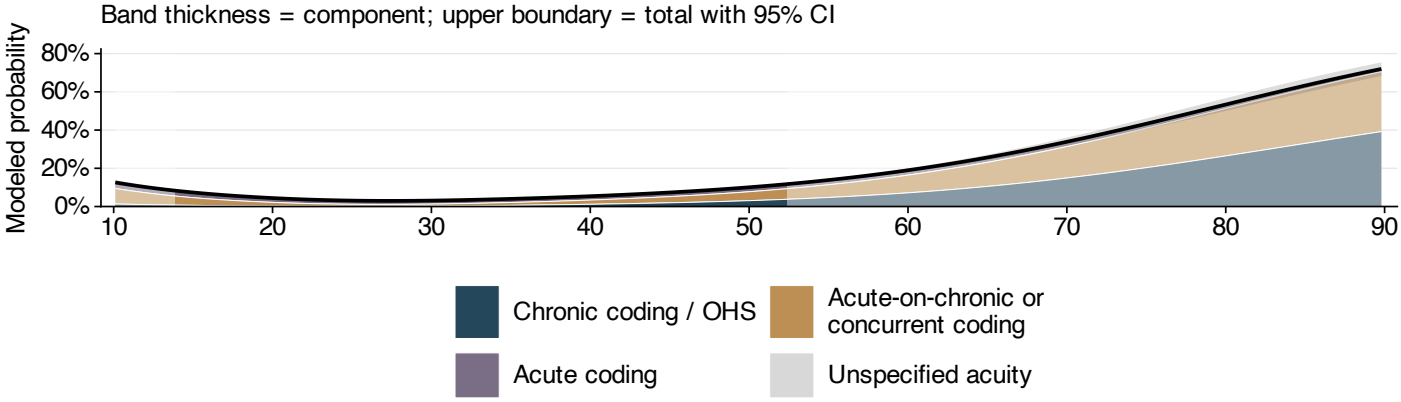
New results: investigator interpretation and figure approval pending.

BMI and documented respiratory-failure acuity / OHS

All valid BMI: primary | Encounter-wide; model/reference N = 380,206 | Firth bias reduction
Events: chronic/OHS 2,198; acute/chronic 6,258; acute 7,339; unspecified 268

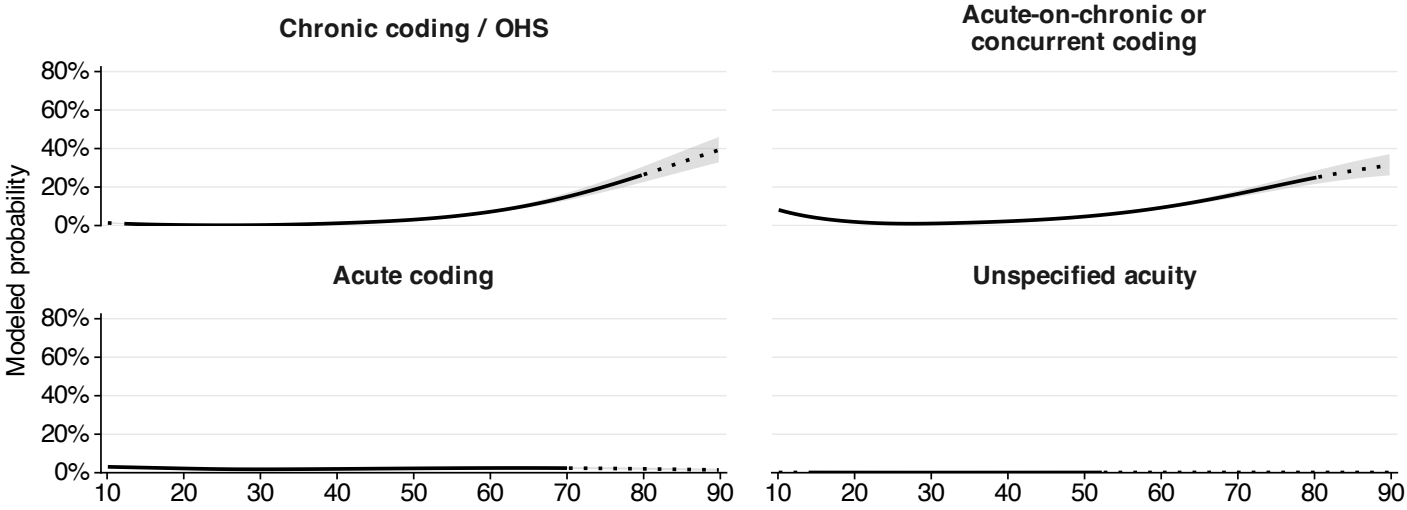
A Documented diagnoses divided by acuity coding

Band thickness = component; upper boundary = total with 95% CI

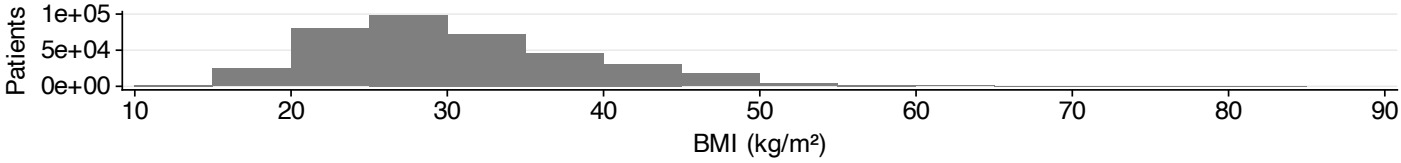


B Component magnitudes on common axes

Pointwise 95% intervals; dotted segments indicate local sparsity



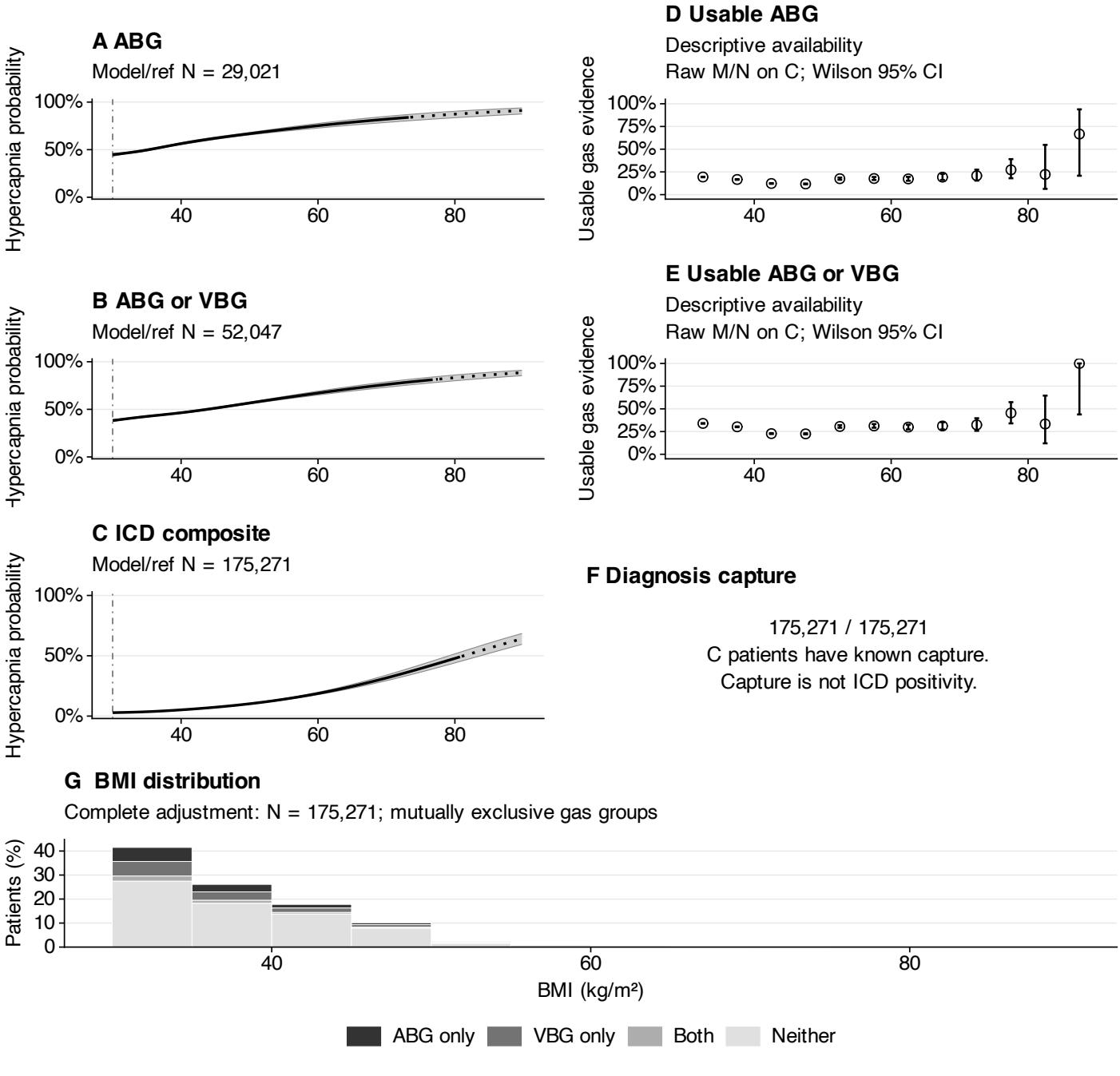
C Patients in the shared diagnosis-capture population



primary/S2-ICD-acuity
All valid BMI: primary. Mean-bias-reduced five-category multinomial spline; all diagnosis-evaluable patients in the stated population retained. Encounter-wide coding, no gas requirement; inherited adjustment and the population-specific spline basis. Bands partition the composite; thickness gives component probability. Shading: pointwise 95% intervals from joint information covariance, conditional on the empirical reference and basis. Fading/dots: local sparsity; histogram N equals model/reference N. Acute-on-chronic includes explicit or concurrent acute/chronic coding. OHS (obesity hypoventilation syndrome) is analytically chronic; this does not establish chronic respiratory failure. Source selection and diagnostic incorporation limit physiological interpretation.
New results: investigator interpretation and figure approval pending.

BMI and documented hypercapnia

BMI ≥30: secondary | encounter-wide



primary_bmi_ge30/F1-B

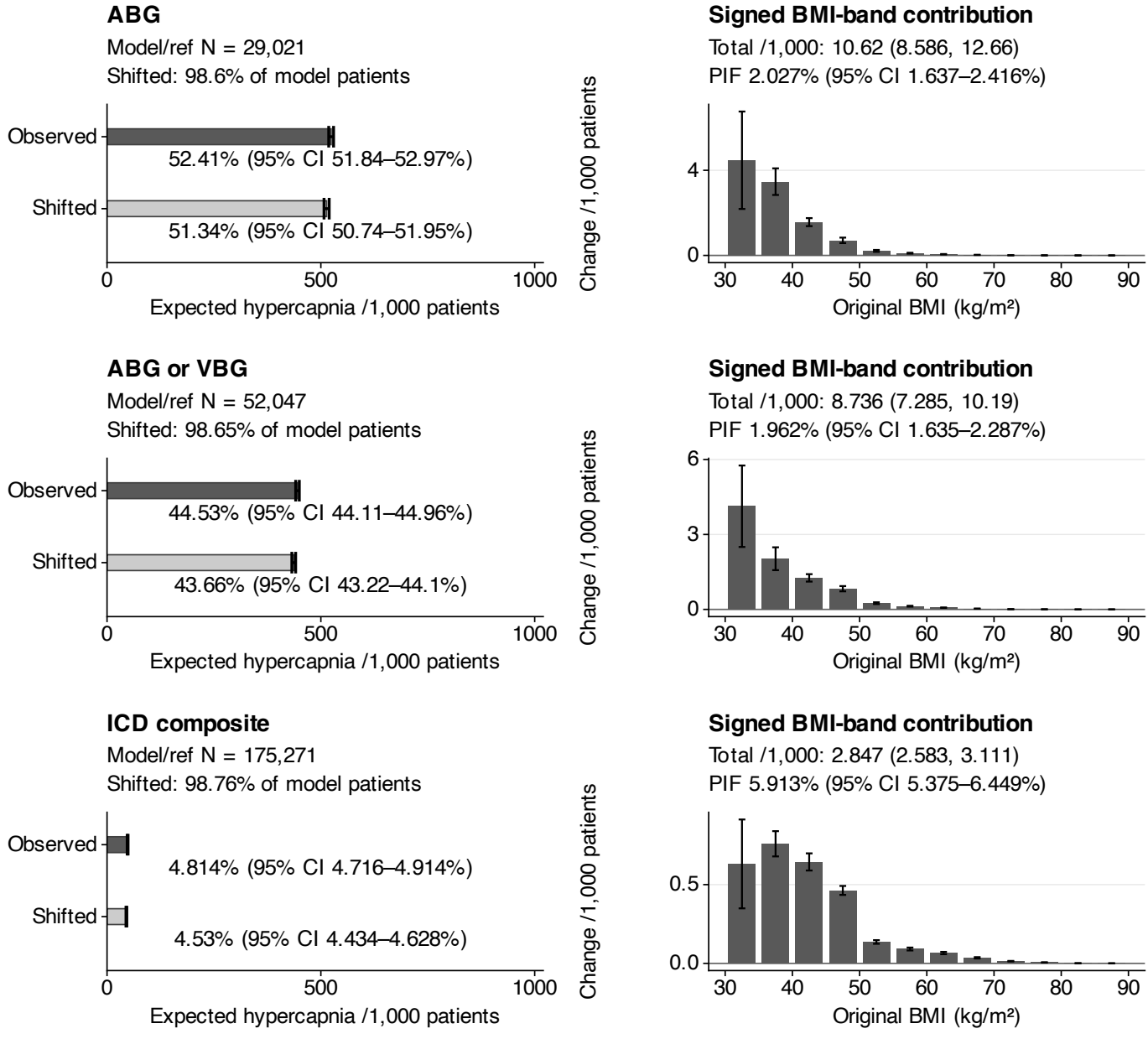
Analysis population: BMI ≥30: secondary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F1. BMI–hypercapnia relationships at the first eligible adult 2022 acute-care encounter. ABG denotes $PaCO_2$ ≥45 mm Hg; ABG or VBG denotes $PaCO_2$ ≥45 or $PvCO_2$ ≥50 mm Hg. ICD denotes the five-code composite, including E66.2. The vertical dot-dash line marks BMI 30, the scenario eligibility boundary and this secondary population's BMI cutoff. Gas curves standardize over outcome-evaluable patients; ICD uses known diagnosis capture. Model/ref N is the common model and reference count. Availability and BMI frequency use all adjustment-complete patients (C); the four frequency groups are mutually exclusive. Adjustment: age, sex, race/ethnicity and region with a fixed natural BMI spline. Shading and fine boundaries show pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. Open availability points use Wilson intervals when no model is estimable. Known capture is distinct from ICD positivity. ABG/VBG: arterial/venous blood gas. Conditional model uncertainty excludes selection and measurement assumptions; no population transport or causal effect is inferred.

New results: investigator interpretation and figure approval pending.

One-unit BMI reduction for observed BMI >30; no floor

Paired predictions; original reference and covariates retained

BMI ≥30: secondary | encounter-wide



primary_bmi_ge30/F2

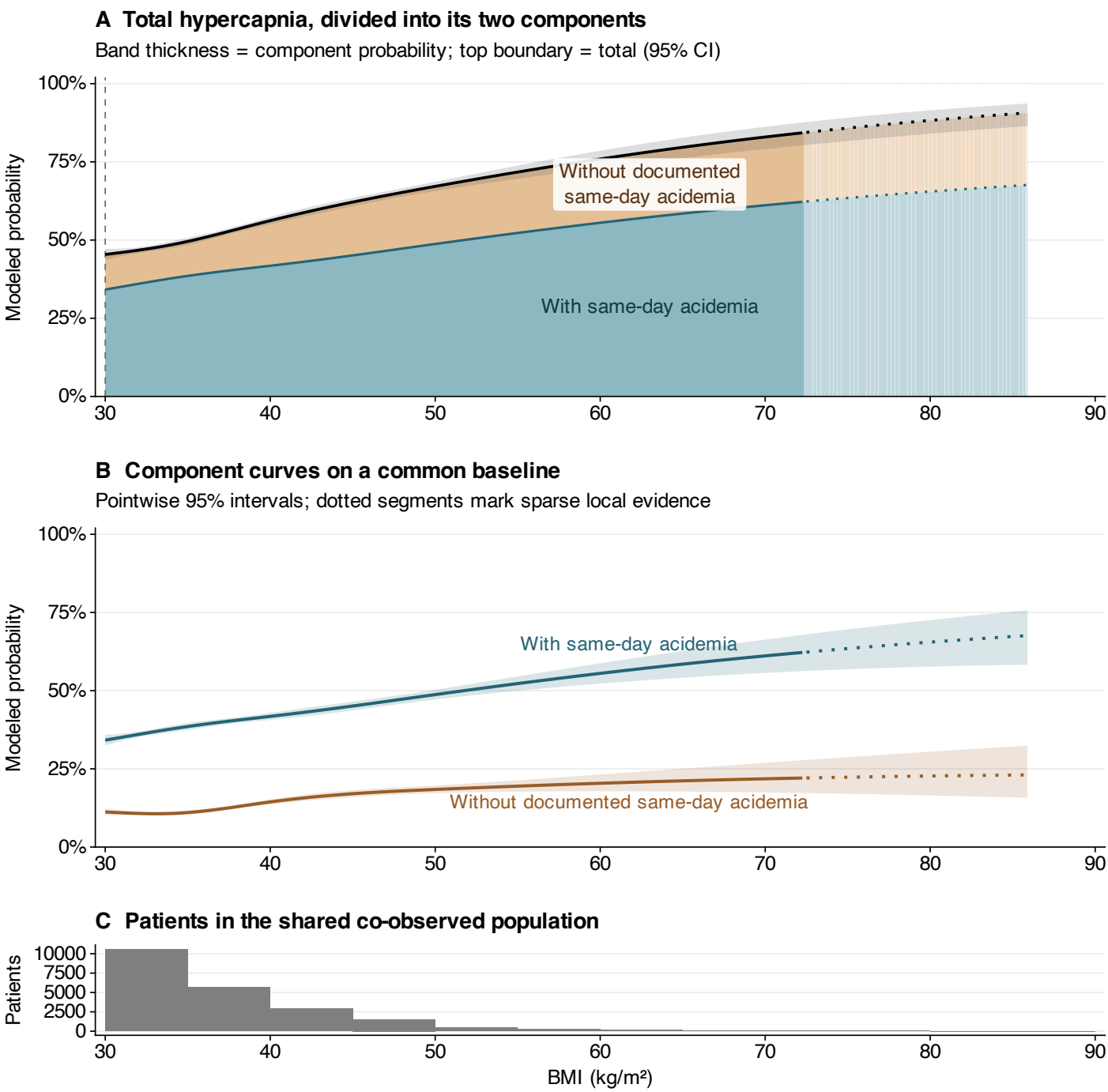
Analysis population: BMI ≥30: secondary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F2. Paired scenario: One-unit BMI reduction for observed BMI >30; no floor. ABG: arterial PCO_2 ≥45 mm Hg; ABG or VBG: arterial ≥45 or venous ≥50 mm Hg; ICD: five-code composite including E66.2. Left: observed and shifted standardized probabilities with model-based 95% CIs and printed limits. Left axes share the full 0–1,000 reference-patient scale; printed probabilities and limits retain small differences. N identifies the common model/reference population; shifted percentage is a descriptive cohort fraction. Right: signed original-BMI-band contributions per 1,000 reference patients, with 95% CIs. Patients at BMI ≤30 remain in baseline burden and contribute exactly zero scenario difference; the eligible share is printed. Total-difference/PIF limits are printed; contribution covariance reproduces total variance. Right axes vary by outcome and retain zero and all limits. Gas samples are outcome-evaluable; ICD uses known capture. Adjustment, source-entry membership and covariates are fixed. PIF is the model-standardized potential impact fraction. These conditional scenarios do not estimate a weight-loss treatment effect.

New results: investigator interpretation and figure approval pending.

BMI and hypercapnia with and without same-day acidemia

Same recorded-day co-occurrence; not verified same-sample pairing

BMI ≥30: secondary; model/ref N = 21,748; with acidemia: 8,639; without: 2,784



primary_bmi_ge30/S1-same-day-ABG

BMI ≥30: secondary. Same recorded-day co-occurrence; not verified same-sample pairing. Fixed-index patients with both arterial analytes share inherited adjustment and their analysis population's fixed BMI spline. Any eligible day with $PaCO_2$ ≥45 mm Hg and pH <7.35 assigns the with-acidemia category; remaining hypercapnic patients have no documented same-day acidemia. Different samples may contribute. A: band thicknesses give component probabilities; the upper boundary is total hypercapnia. B: components share a baseline. Shading denotes pointwise 95% intervals using full joint covariance. C: shared-population BMI frequencies. Fading/dotted lines mark local windows below 30 patients or five events/non-events; marginal checks do not establish joint support. These recorded outcomes do not establish simultaneous physiology, acute respiratory failure or compensation. Selection, treatment timing and confounding remain limitations.

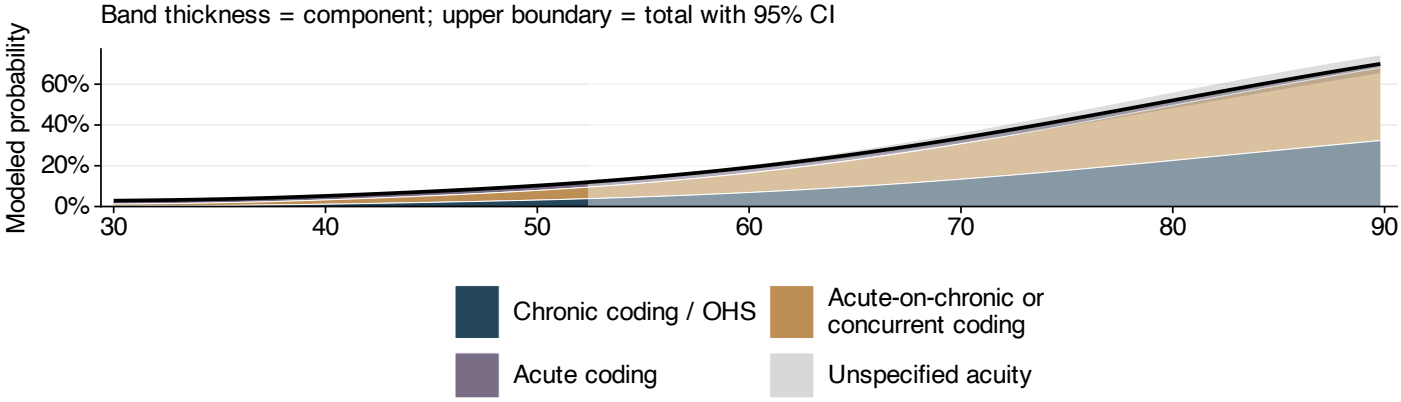
New results: investigator interpretation and figure approval pending.

BMI and documented respiratory-failure acuity / OHS

BMI ≥30: secondary | Encounter-wide; model/reference N = 175,271 | Firth bias reduction
Events: chronic/OHS 1,752; acute/chronic 3,361; acute 3,201; unspecified 124

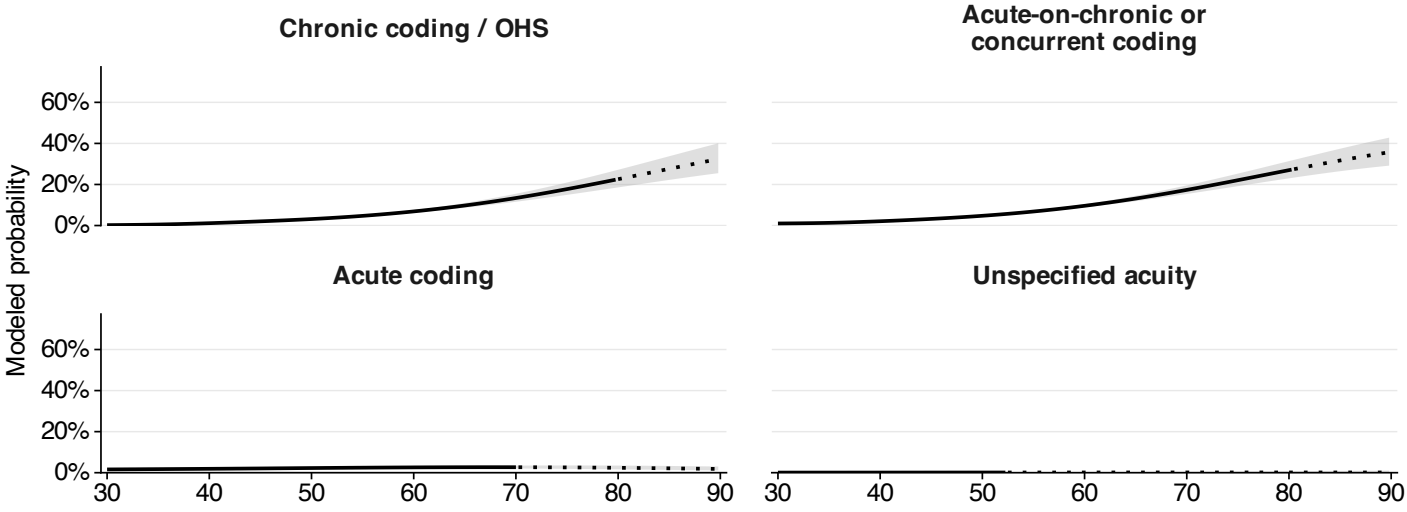
A Documented diagnoses divided by acuity coding

Band thickness = component; upper boundary = total with 95% CI

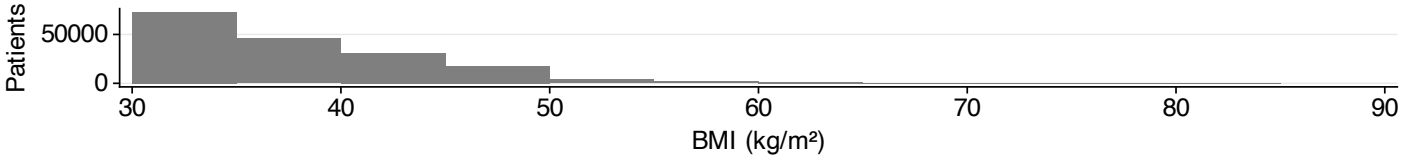


B Component magnitudes on common axes

Pointwise 95% intervals; dotted segments indicate local sparsity



C Patients in the shared diagnosis-capture population



primary_bmi_ge30/S2-ICD-acuity
BMI ≥30: secondary. Mean-bias-reduced five-category multinomial spline; all diagnosis-evaluable patients in the stated population retained. Encounter-wide coding, no gas requirement; inherited adjustment and the population-specific spline basis. Bands partition the composite; thickness gives component probability. Shading: pointwise 95% intervals from joint information covariance, conditional on the empirical reference and basis. Fading/dots: local sparsity; histogram N equals model/reference N. Acute-on-chronic includes explicit or concurrent acute/chronic coding. OHS (obesity hypoventilation syndrome) is analytically chronic; this does not establish chronic respiratory failure. Source selection and diagnostic incorporation limit physiological interpretation.
New results: investigator interpretation and figure approval pending.